Resources

MATH 107: Introduction to Linear Algebra

Daily reading assignments are listed with each meeting on Lectures.

Textbook

The main text is Stephen Boyd and Lieven Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (VMLS). The book is free to read and download from the authors.

Boyd lecture videos: Stanford ENGR108 (2020) recordings that follow VMLS are on YouTube: full playlist. When a MATH 107 meeting has a matching chapter, the corresponding video link appears at the top of that day's lecture notes (HTML and PDF).

You do **not** need to buy Gilbert Strang, *Introduction to Linear Algebra*. A few meetings use three freely posted MIT files. The ILA, 6th edition resource page distinguishes those sample sections from the commercial book.

Strang lecture videos: Gilbert Strang's MIT 18.06 recordings are on YouTube (playlist). Matching free PDF + video links appear at the top of the relevant lecture notes (HTML and PDF).

When	File	What to read	Matching video (optional)
Nov 4	§3.5 Dimensions of the Four Subspaces	Rank, column space, null space, basis, and dimension. Not the whole section: row space and left nullspace are enrichment, not quiz material.	Lec 6, Lec 10
Nov 16–20	§6.1 Introduction to Eigenvalues	$Ax = \lambda x$, dynamics, the characteristic equation, and the 90° rotation with eigenvalues i and $-i$. This PDF is only §6.1; later Chapter 6 sections listed at the start of the file are not free from this folder.	Lec 21, Lec 22
Dec 4	SVD slides	Selected slides (not all 17 pages) on SVD, low-rank approximation, data compression, least squares, and image processing.	Lec 29 SVD, short SVD

Software

We use **Octave** or **Python** (pick one for the term). Each lab hub has exercises in both languages. Credit is an in-class one-pager with instructor initials — not a Canvas file upload (see Lab 0). Lab 0 assumes no prior coding. Bring a laptop to lab days, or let me know at least a day ahead to bring a campus one for you. **Labs meet on Fridays** (same rule as quizzes).

Growing cheat sheet: Octave / Python syntax reference.

Primary Octave reference: Jason Lachniet, *Introduction to GNU Octave: A Brief Tutorial for Linear Algebra and Calculus Students* (3rd ed., 2020). Free PDF, licensed CC BY-SA 4.0. Author, title, and PDF access are also listed at the Open Textbook Library.

Use Lachniet for syntax, commands, and selected exercises. The **lab hub pages** are the assigned work (Octave and Python on the same page). Daily chapter pointers are on Lectures and each lab; it is all one Lachniet PDF, not a new link each lab.

Lab	Lachniet chapters
Lab 0	Ch. 1 Getting Started — interface, variables, vectors/matrices, basic operations, plotting
Lab 1	Ch. 1 — vectors, matrix storage, indexing, plots (plus our distance/nearest-neighbor scaffold)
Lab 2	Ch. 1 — distances, nearest neighbors, scale (plus syntax sheet)
Lab 3	Ch. 2 — matrices and linear-algebra computations (plus our transformation plotting code)
Systems / RREF (Oct 14–16)	Ch. 2 — solving systems, Gaussian elimination/RREF, linear-system commands
Lab 4	Ch. 2 matrix operations, plus loops/programming from the later introductory programming portions
Lab 5	Ch. 5 — eigenvalues/eigenvectors, Markov chains, diagonalization
Lab 6	Ch. 2, especially least-squares / data-fitting exercises
Capstone / Lab 7	Ch. 5 SVD, plus Ch. 7 project on SVD / image compression

Optional reference: the GNU Octave documentation (not assigned reading). Download the program from octave.org.

Python / Google Colab

If you choose Python, use Google Colab for lab work (hosted Jupyter; no local install). Exercises appear on each lab hub next to the Octave versions.

Role	Tool	Link
Where we write and run Python	Google Colab	colab.google
Organize and prepare tabular data	Polars	Getting started
Vectors, matrices, and linear algebra	NumPy	Beginner guide; later, linear algebra
Visualize and communicate results	Plotnine	Introductory guide

Polars is for real tables: reading files, selecting columns, filtering, transforming, and handing numerical columns to NumPy. **NumPy** is for arrays, indexing, and later `solve`, `eig`, `lstsq`, `svd`, norms, and determinants. **Plotnine** is

for scatterplots, line plots, fitted relationships, residual plots, clustering graphics, and other standard figures; it uses a grammar-of-graphics syntax and works directly with Polars DataFrames.

As with Lachniet, these pages support syntax. The lab hub and in-class one-pager are the assigned work.

Campus

- Cal Poly Humboldt
- Library
- Student Disability Resource Center
- Counseling and Psychological Services

Tutoring and drop-in help: check the current campus learning-center listings once the term site is posted, or ask in office hours.

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